CAP6412
Advanced Computer Vision
Spring 2026

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UCF Global 223

Diffusion Model

Lecture 6

Generative AI

- **Text**
  - Large Language Models (LLM)
    - Generate/Predict the next word
    - Auto-regressive
  - Visual Language Models (VLM)/Large Multimodal Models (LMM)

- **Images**
  - DALL-E
    - Auto-regressive model for image Generation
  - Diffusion Models
    - Generate Images, Videos, audio, …

Generative AI

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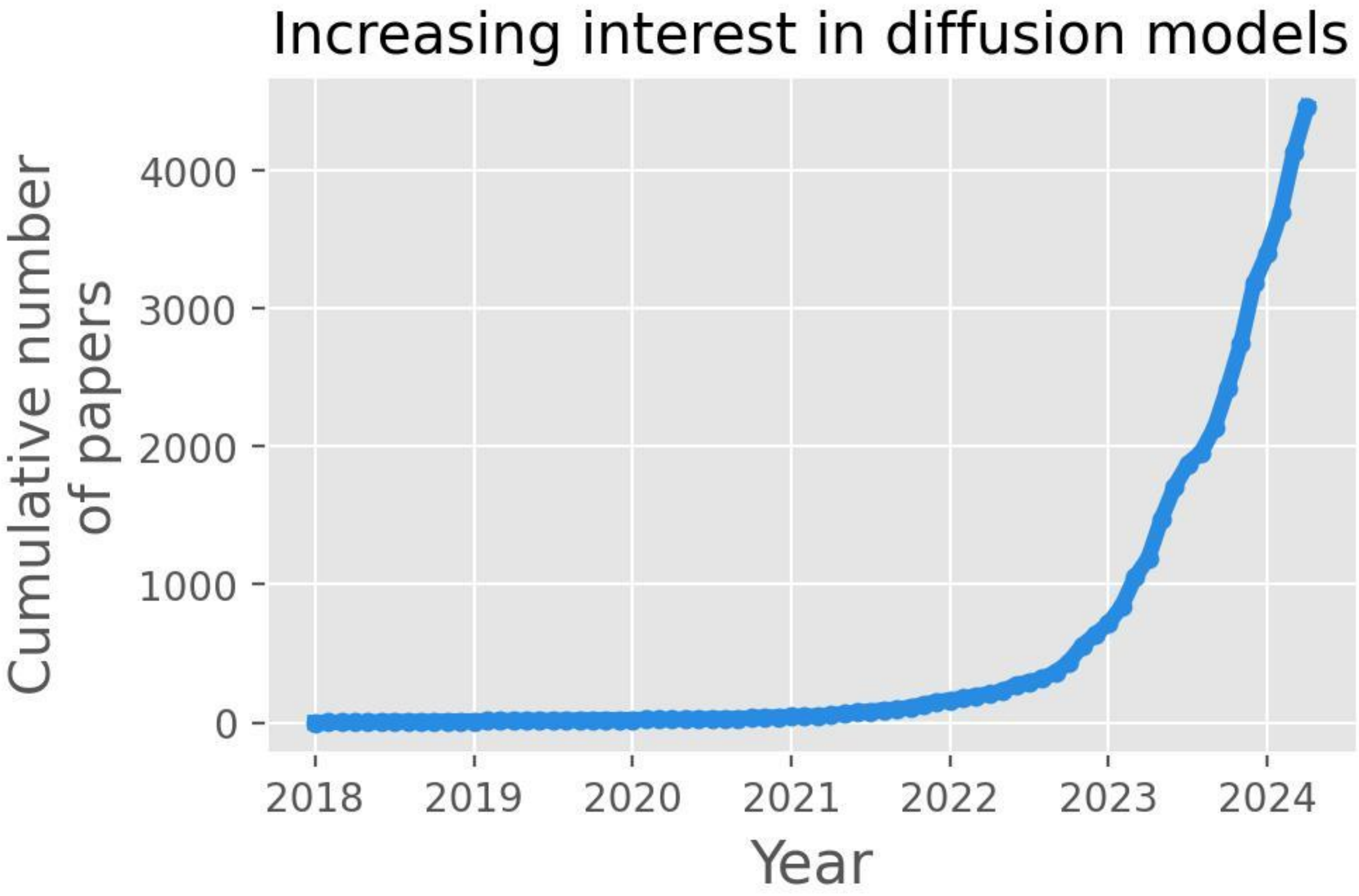
- **Images**
  - DALL-E
    - Auto-regressive model for image Generation
  - **Diffusion Models**
    - **Generate Images**, Videos, audio, …

Motivation


Increasing interest in diffusion models
Cumulative number of papers
4000
3000
2000
1000
0
2018
2019
2020
2021
2022
2023
2024
Year

Diffusion Applications

- Image generation
- Text-to-image
- Super resolution
- Image editing
- Inpainting
- Image segmentation
- Video Generation
- …..

Image generation

A hedgehog using a calculator.

A corgi wearing a red bowtie and a purple party hat.

A transparent sculpture- of a duck made out of glass.

A photo of a Corgi dog riding a bike in Times Square. It is wearing sunglasses and a beach hat.

Pomeranian king with tiger soldiers.

Zebras roaming in the field.

Super resolution

256 x 256 samples on CelebA-HQ.

Inpainting

Image Colorization

Image Colorizations

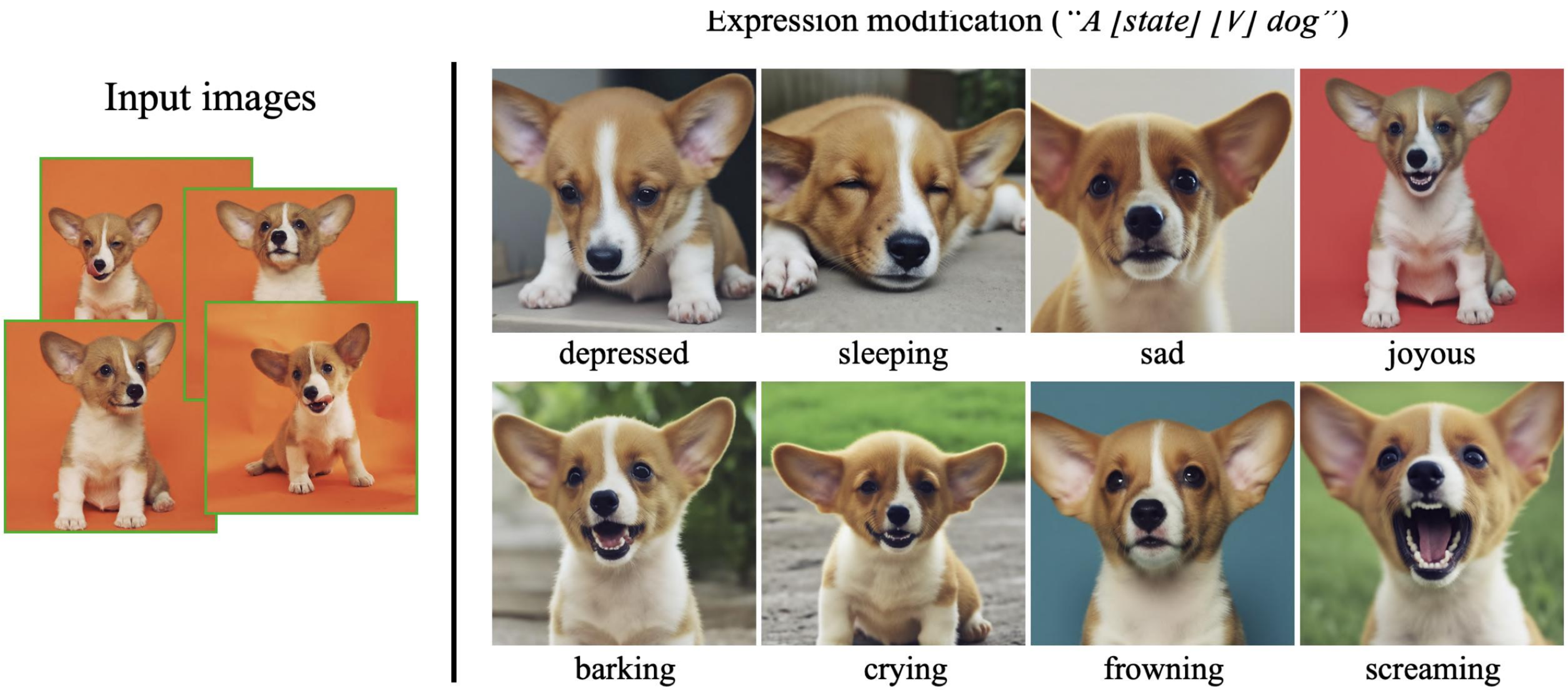
We can even colorize gray-scale portrays of famous people in history (Abraham Lincoln) with a time-dependent score-based model trained on FFHQ. The image resolution is 1024 x 1024.


Input images
in the Acropolis
swimming
sleeping
in a doghouse
in a bucket
getting a haircut
Input images
worn by a bear
in the jungle
on red fabric
at Mt. Fuji
on top of snow
with Eiffel Tower

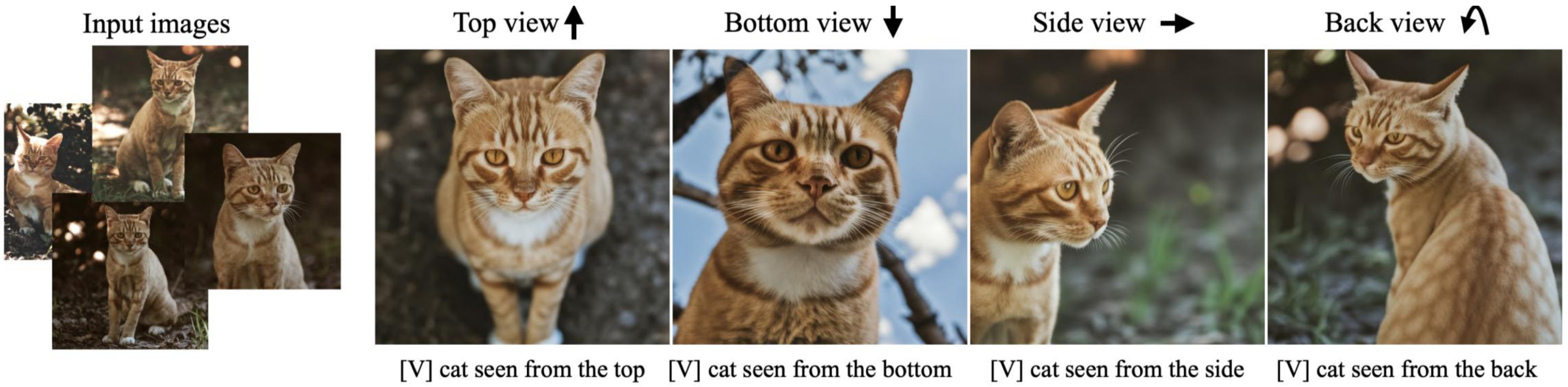
Style transfer

Input images

Expression change



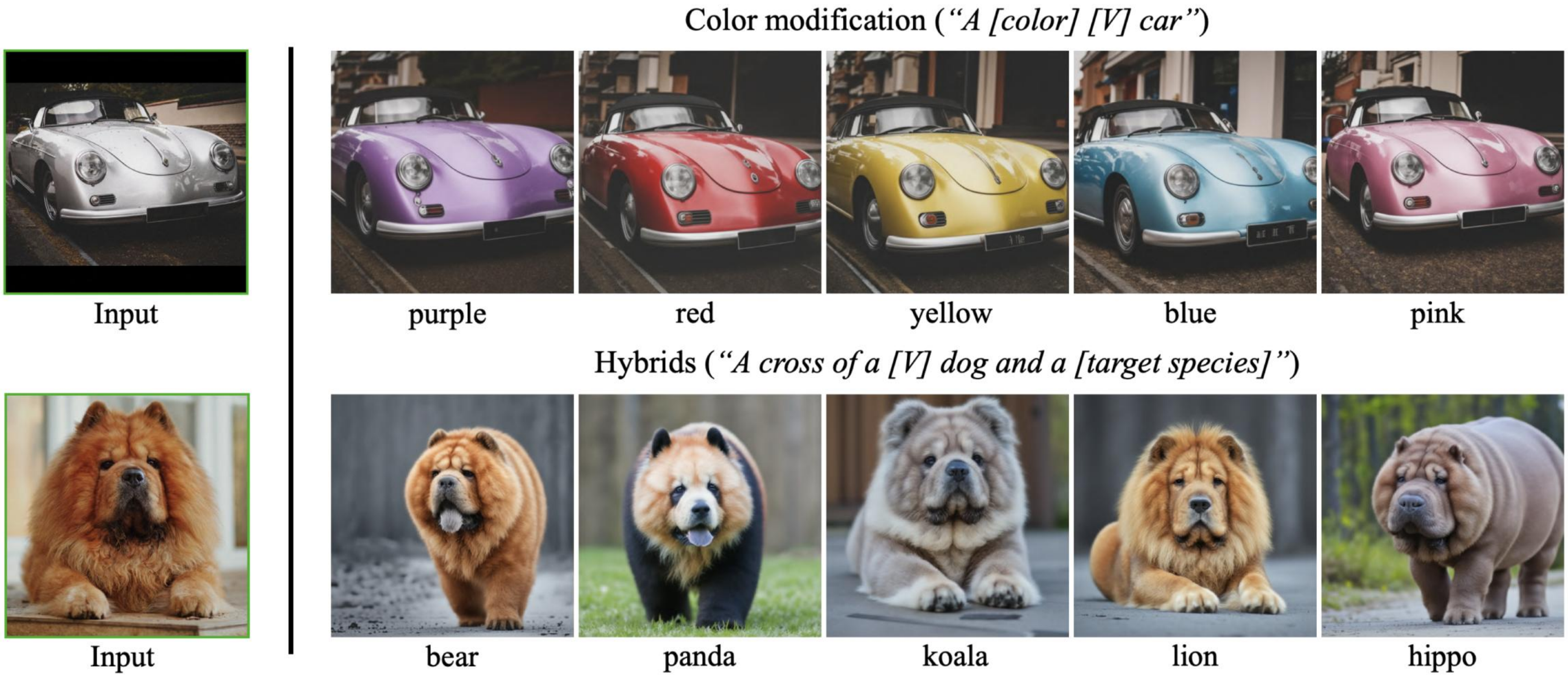
Viewpoint


Input images
Top view
Bottom view
Side view
Back view
[V] cat seen from the top
[V] cat seen from the bottom
[V] cat seen from the side
[V] cat seen from the back

Attributes


Input images
Chef Outfit
Witch Outfit
Ironman Outfit
Nurse Outfit
Purple Wizard Outfit
Superman Outfit
Police Outfit
Angel Wings

Attributes


Color modification ("A [color] [V] car")
Input
purple
red
yellow
blue
pink
Hybrids ("A cross of a [V] dog and a [target species]")
Input
bear
panda
koala
lion
hippo

Video Generation: OpenAI Sora

Several giant wooly mammoths approach treading through a snowy meadow, their long wooly fur lightly blows in the wind as they walk, snow covered trees and dramatic snow capped mountains in the distance, mid afternoon light with wispy clouds and a sun high in the distance creates a warm glow, the low camera view is stunning capturing the large furry mammal with beautiful photography, depth of field.

Source: https://openai.com/index/sora/

Video Generation: Google Veo2

A low-angle shot captures a flock of pink flamingos gracefully wading in a lush, tranquil lagoon. The vibrant pink of their plumage contrasts beautifully with the verdant green of the surrounding vegetation and the crystal-clear turquoise water. Sunlight glints off the water's surface, creating shimmering reflections that dance on the flamingos' feathers. The birds' elegant, curved necks are submerged as they walk through the shallow water, their movements creating gentle ripples that spread across the lagoon. The composition emphasizes the serenity and natural beauty of the scene, highlighting the delicate balance of the ecosystem and the inherent grace of these magnificent birds. The soft, diffused light of early morning bathes the entire scene in a warm, ethereal glow.

Source: https://blog.google/technology/google-labs/video-image-generation-update-december-2024/

Overview

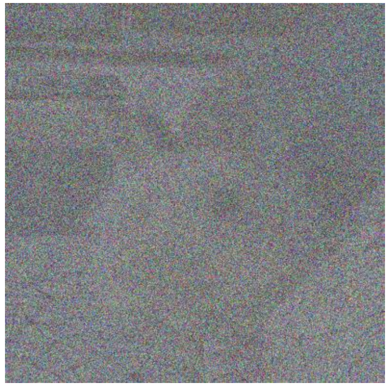
- Diffusion models are probabilistic models
- They involve reversing the process of gradually degrading the data
- Consist of two processes:
  - **The forward process**: data is progressively destroyed by adding noise across multiple time steps
  - **The reverse process**: using a neural network, noise is sequentially removed to obtain the original data

Core idea

Forward process

x_0 x_T

...



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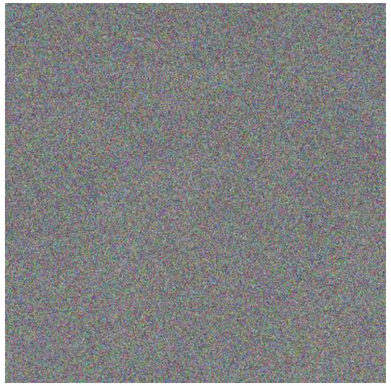


$x_0 \sim q(x)$ $x_T \sim \mathcal{N}(0, I)$

Core idea

x_0 x_T

Reverse process

$x_0 \sim q(x)$ $x_T \sim \mathcal{N}(0, I)$

Forward Diffusion Process

We gradually add Gaussian noise:

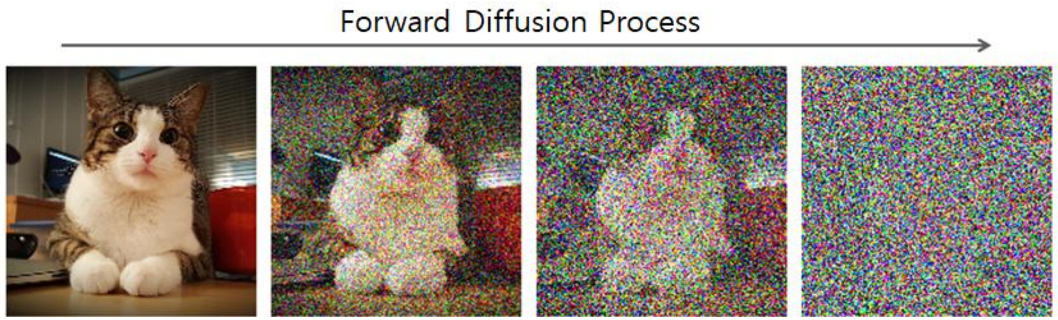
$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I)$$

More explicitly: $x_t = \sqrt{1-\beta_t}\, x_{t-1} + \epsilon$

where, $\epsilon \sim \mathcal{N}(0, \beta_t I)$

Meaning:

- each step slightly corrupts image
- after many steps → pure Gaussian noise



Key point:

- This process is fixed and not learned.

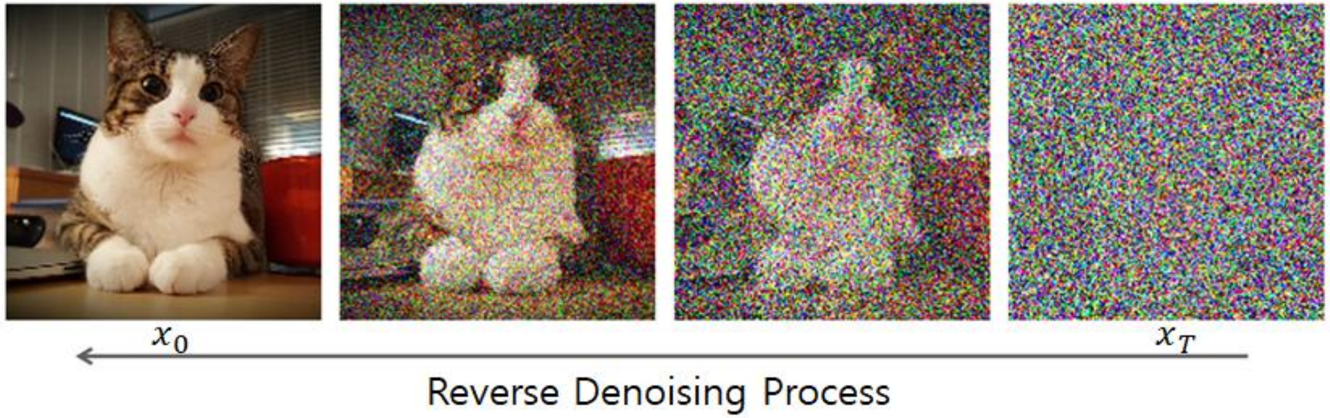
Reverse Process Goal

We want:

$$p_\theta(x_{t-1} \mid x_t)$$

Model learns:

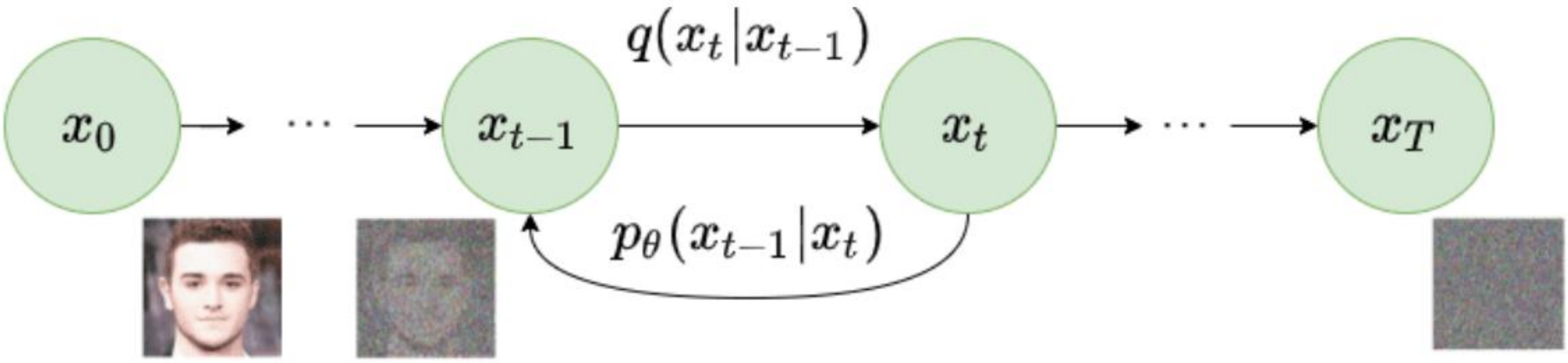
- how to remove noise
- step by step
- until clean image



Key point:

- This process is learned.

Core idea



Reverse diffusion process. Image modified by Ho et al. 2020

What the Network Actually Predicts

- Important simplification:
  - Instead of predicting image, model predicts:

noise ϵ

- Training objective:

$$\mathcal{L} = || \epsilon - \epsilon_\theta(x_t, t) ||^2$$

- Predicting noise turns out easier than predicting pixels.

Training Algorithm

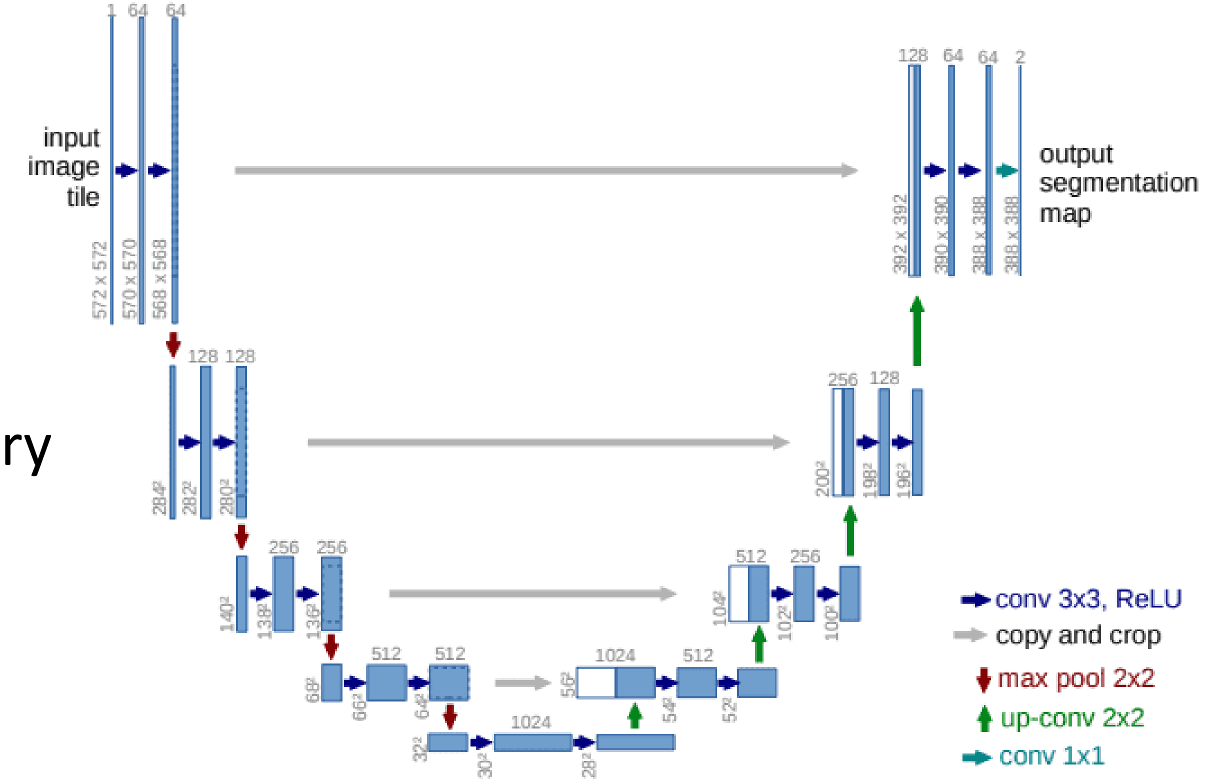
For each training image:

- Sample timestep **t**
- Add noise to image
- Ask model to predict noise
- Compute MSE loss

- No adversarial loss…
- No discriminator…
- Stable training…

Architecture Used

- Most diffusion models use:
  - **U-Net backbone**
- Why?
  - multi-scale
  - preserves spatial structure
  - skip connections help detail recovery
- Input to network:
  - noisy image
  - timestep embedding



Why Diffusion Works So Well

Key reasons:

1. Stable training
(no adversarial game)

2. Likelihood-based
(grounded objective)

3. Progressive refinement
(easier learning task)

4. Scales extremely well

Strengths vs Weaknesses

- **Strengths**
  - very high image quality
  - stable training
  - flexible conditioning
- **Weaknesses**
  - slow sampling
  - expensive compute
  - many steps required

Generative AI (Computer Vision)

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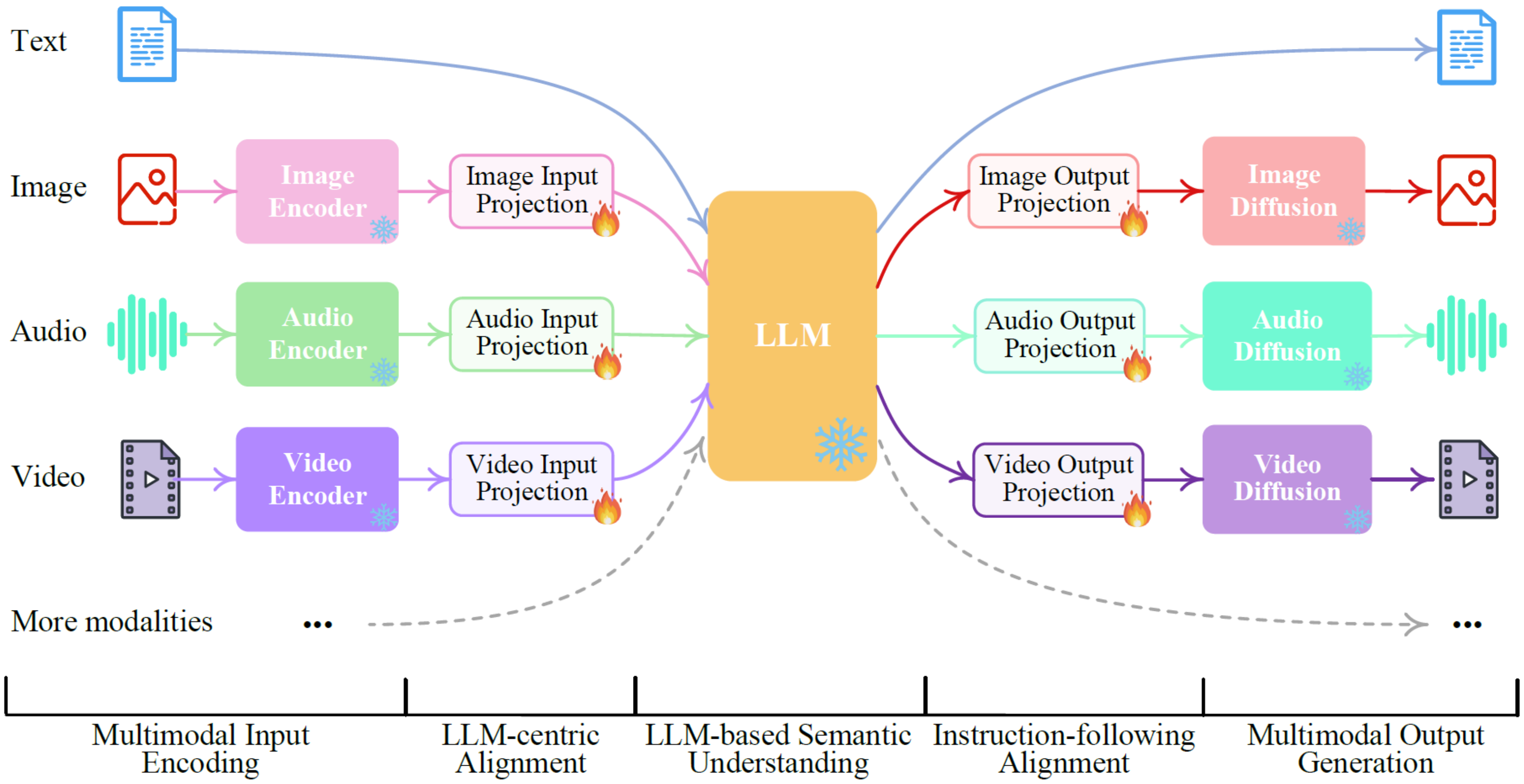
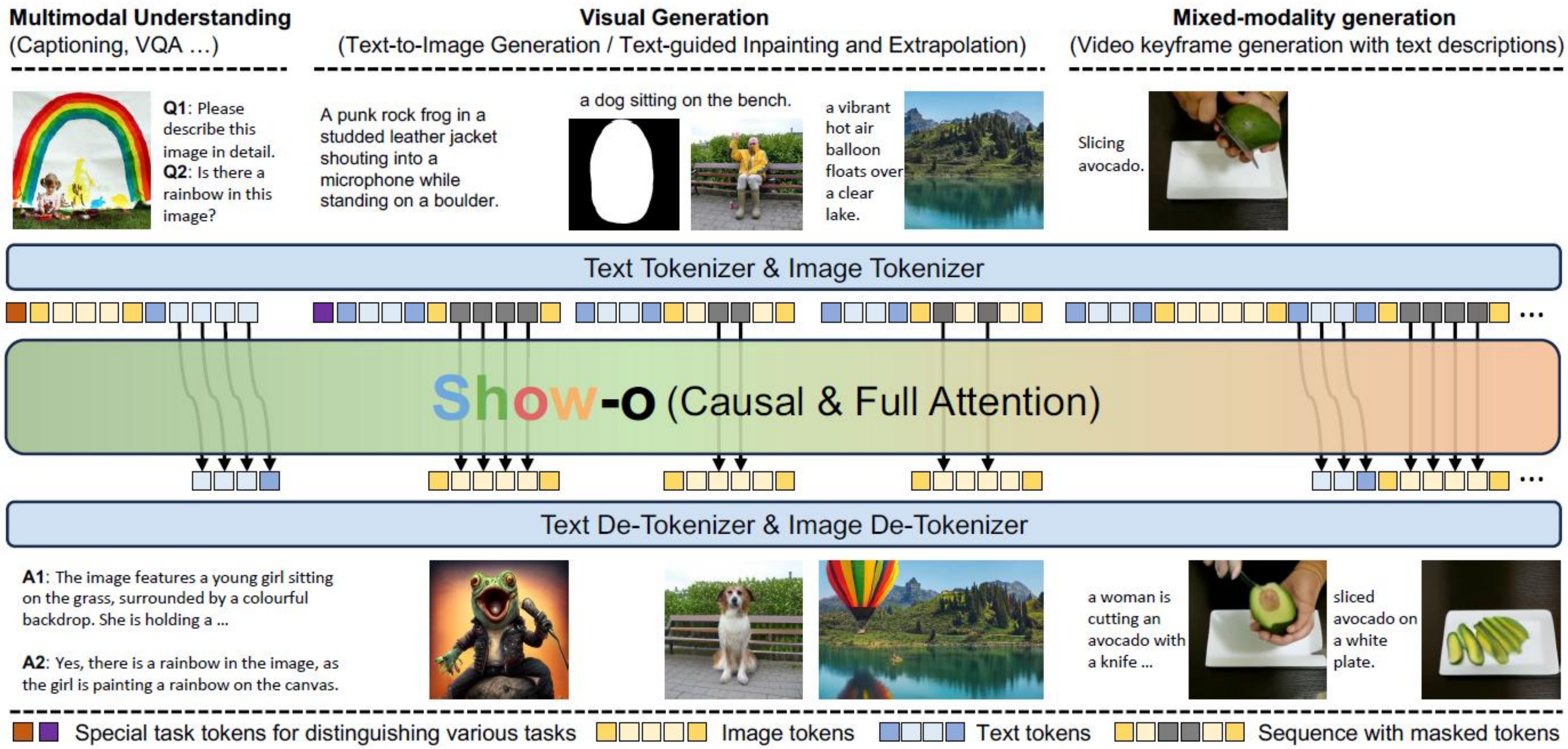
Can we do both Text and Image Generation?



Can we do both using one transformer?



Questions?

Thank you!

Slides credit: Mubarak Shah